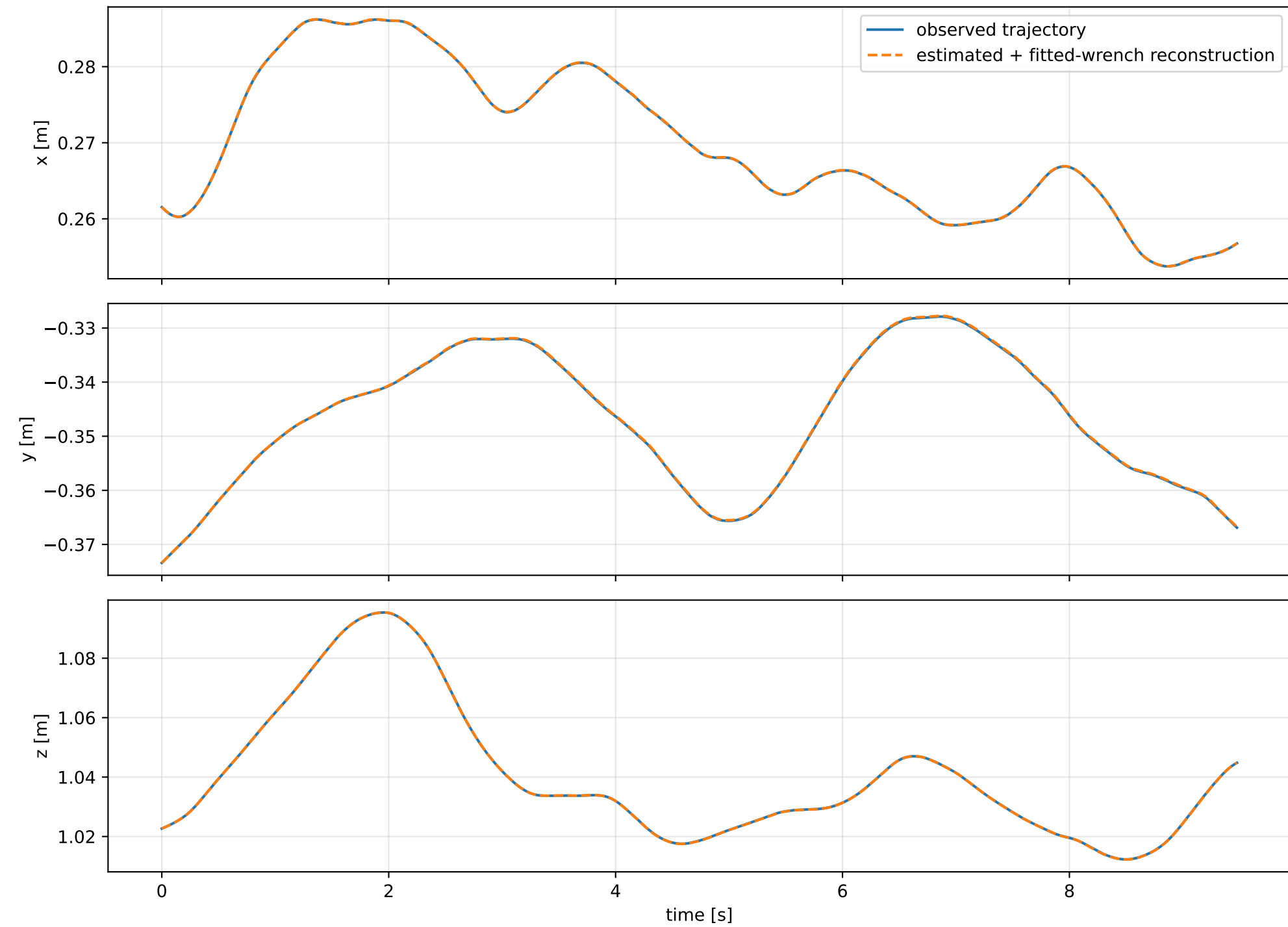
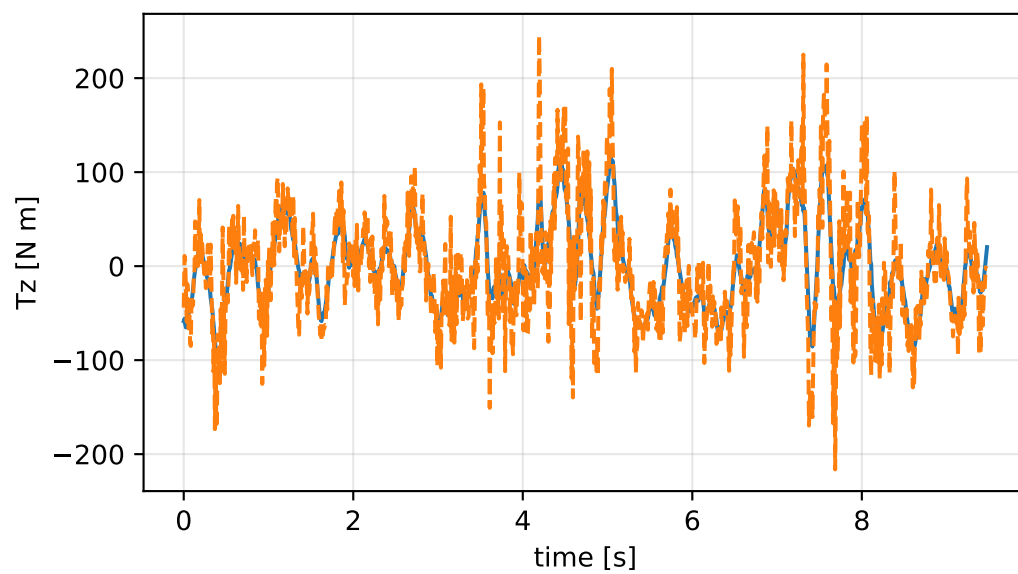
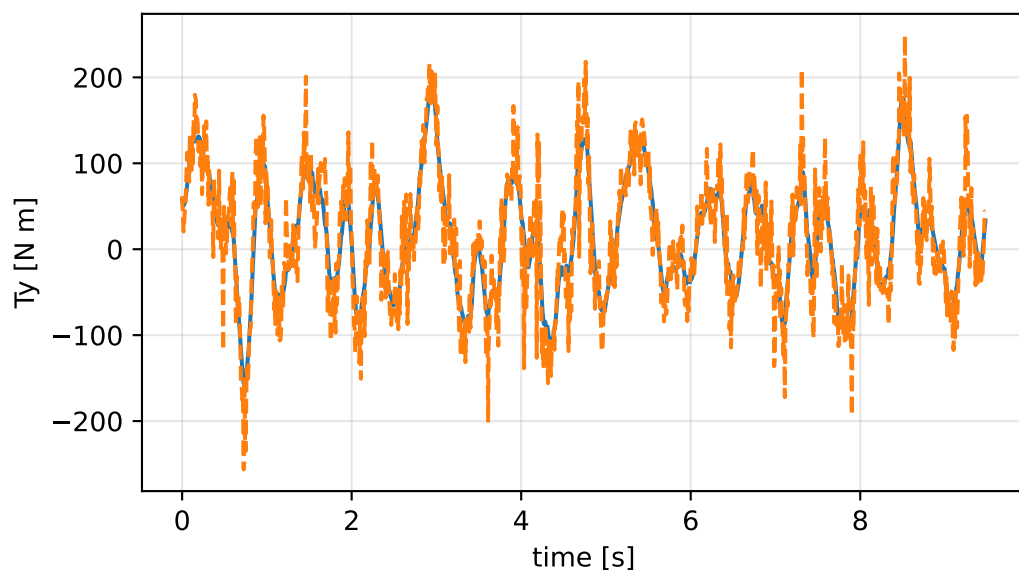
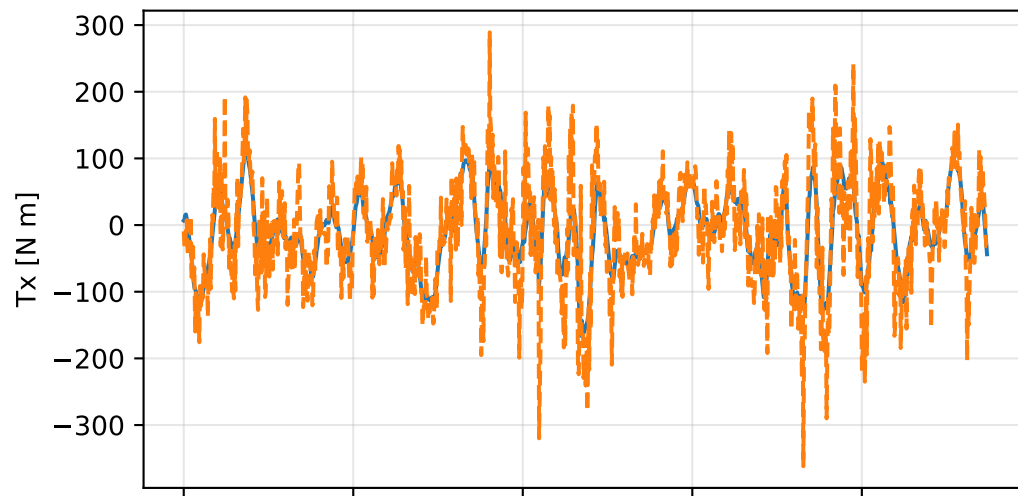
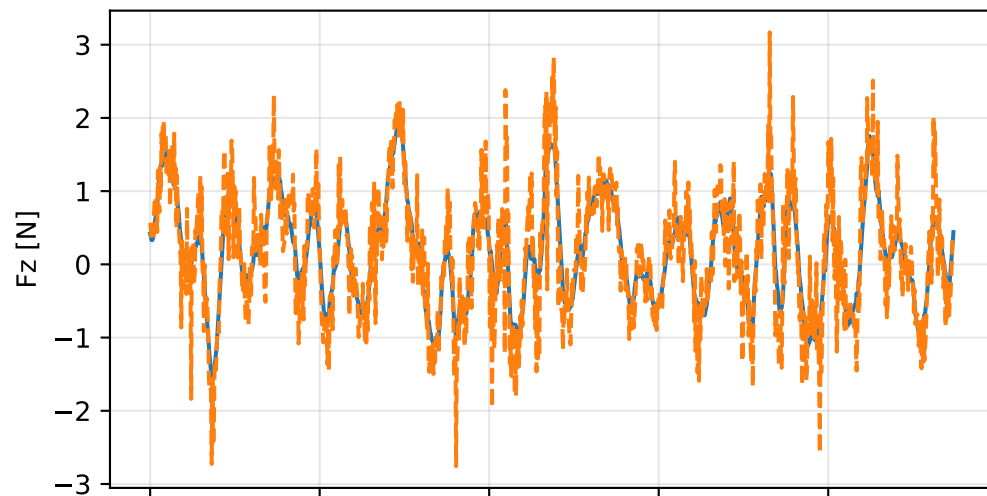
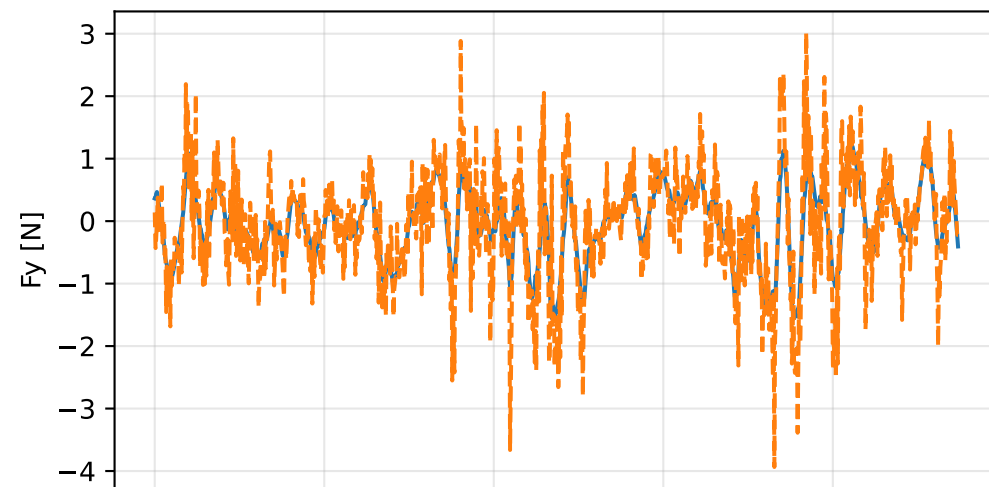
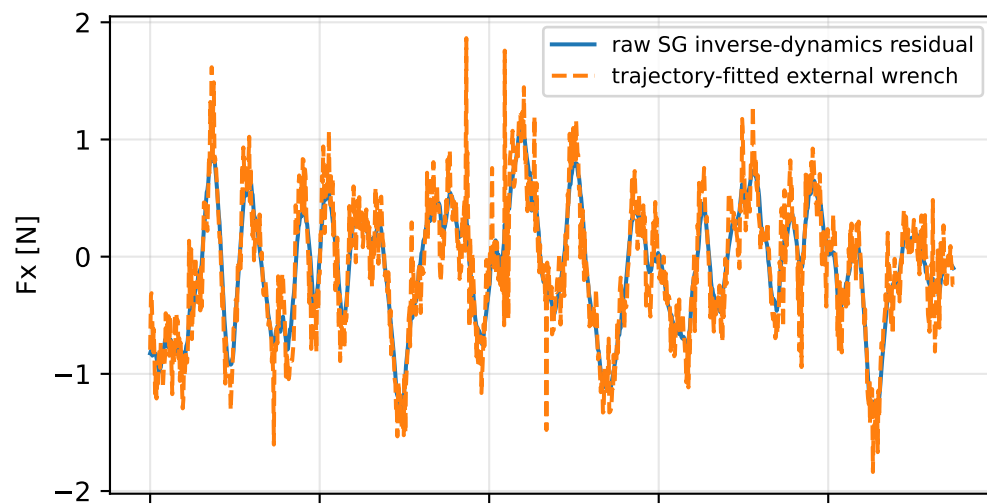


sweep_lag_bounds__000: trajectory

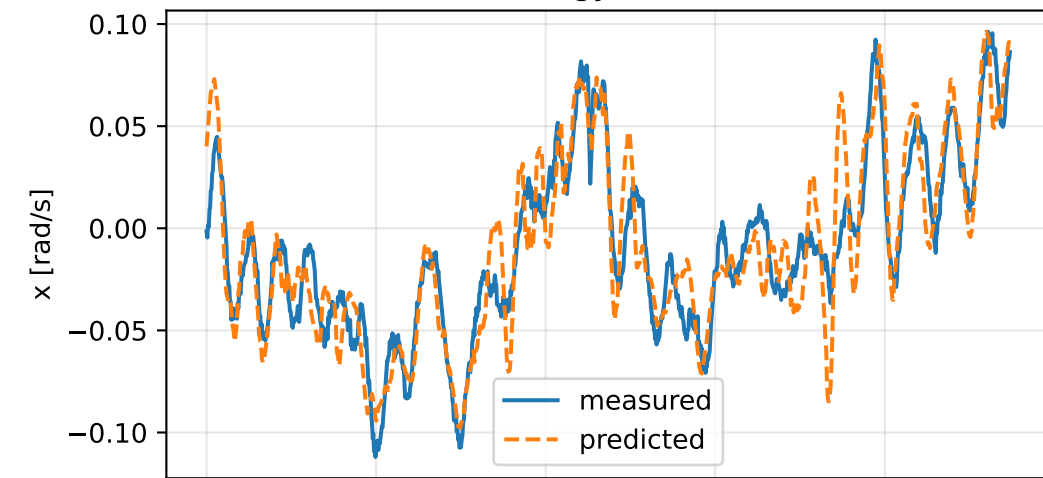


sweep_lag_bounds_000: residual wrench

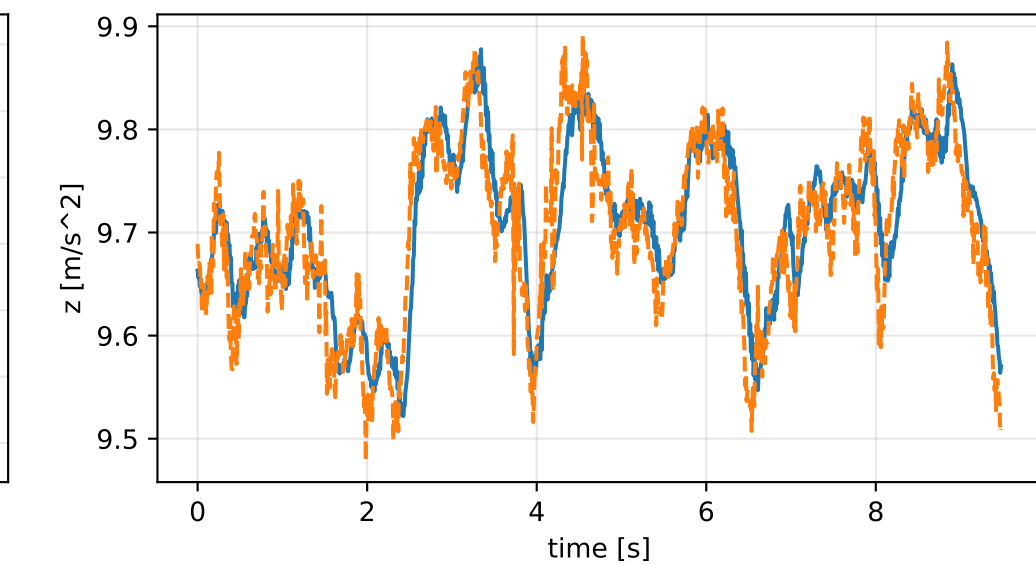
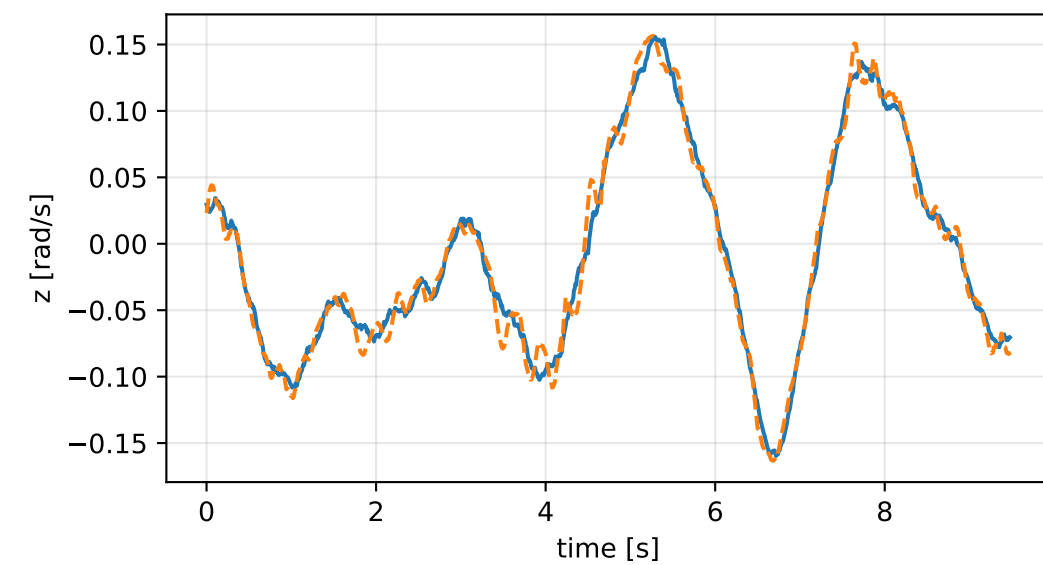
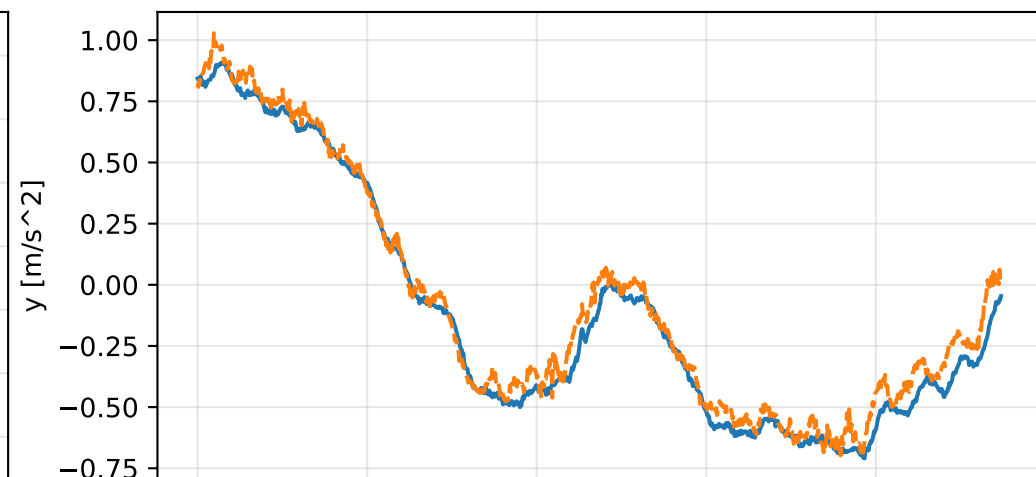
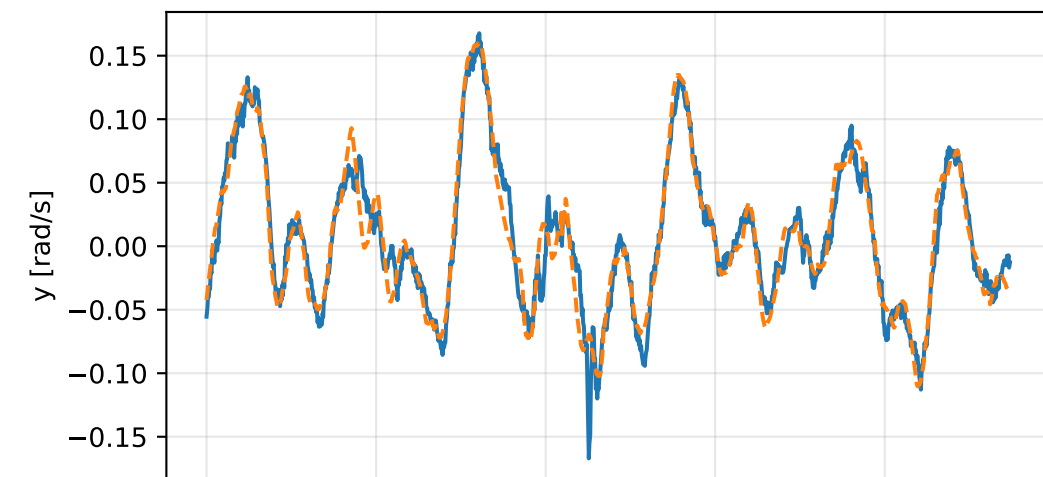
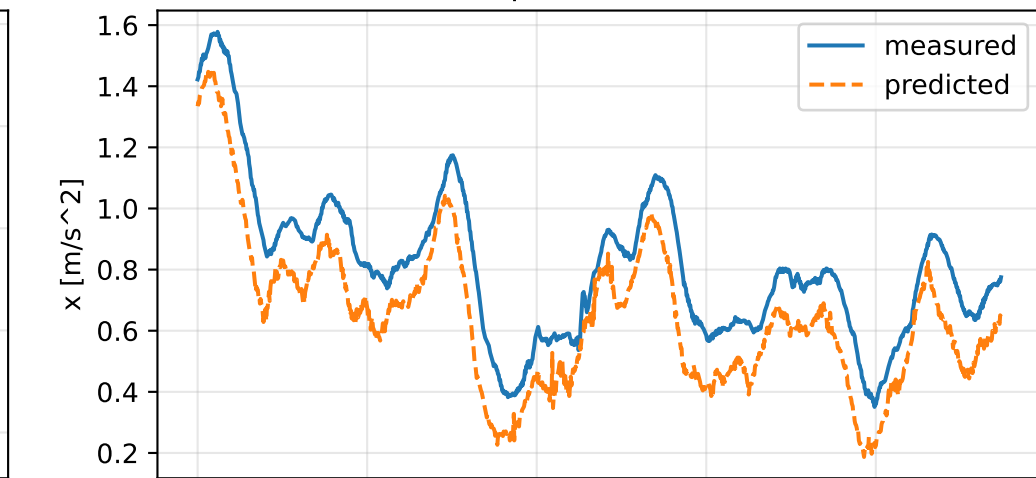


sweep_lag_bounds_000: measured sensor comparison

gyro



specific force



sweep_lag_bounds_000: nominal -> estimated parameters

Case: sweep_lag_bounds_000

status: completed (all stages completed)

mass [kg] nominal 2.35159759 estimated 1.19497974 delta -1.15661785

inertia [kg m^2] (nominal -> estimated; delta)

[6.5000061e-02 -7.2789925e-07 1.9015080e-05] -> [129.3495826 -73.3129721 -81.7775565] d=[129.2845825 -73.3129714 -81.7775565]
[-7.2789925e-07 6.4952656e-02 5.9167305e-08] -> [-73.3129721 165.0176848 -64.1995826] d=[-73.3129714 164.9527322 -64.1995826]
[1.9015080e-05 5.9167305e-08 1.2899211e-01] -> [-81.7775565 -64.1995826 150.9737913] d=[-81.7775565 -64.1995826 150.8447913]

CoG [m] [-2.0247086e-03 -3.0526579e-05 9.5097496e-03] -> [-1.304714 -0.9258715 -1.1299209] d=[-1.3026893 -0.9258414 -1.1299209]

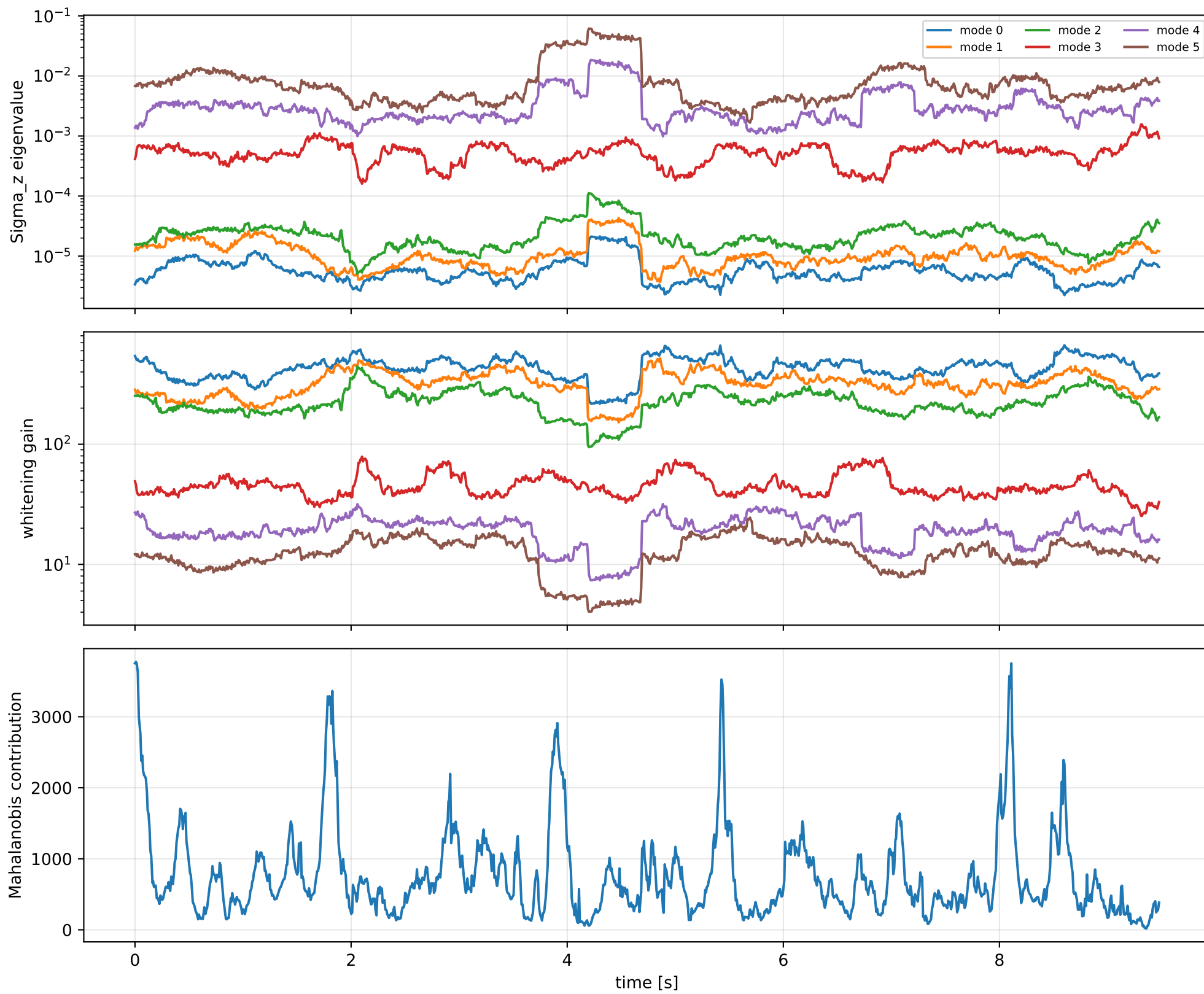
rotor force effectiveness

[1. 1. 1. 1.] -> [0.2535314 0.2667904 0.4609629 0.6306019] d=[-0.7464686 -0.7332096 -0.5390371 -0.3693981]

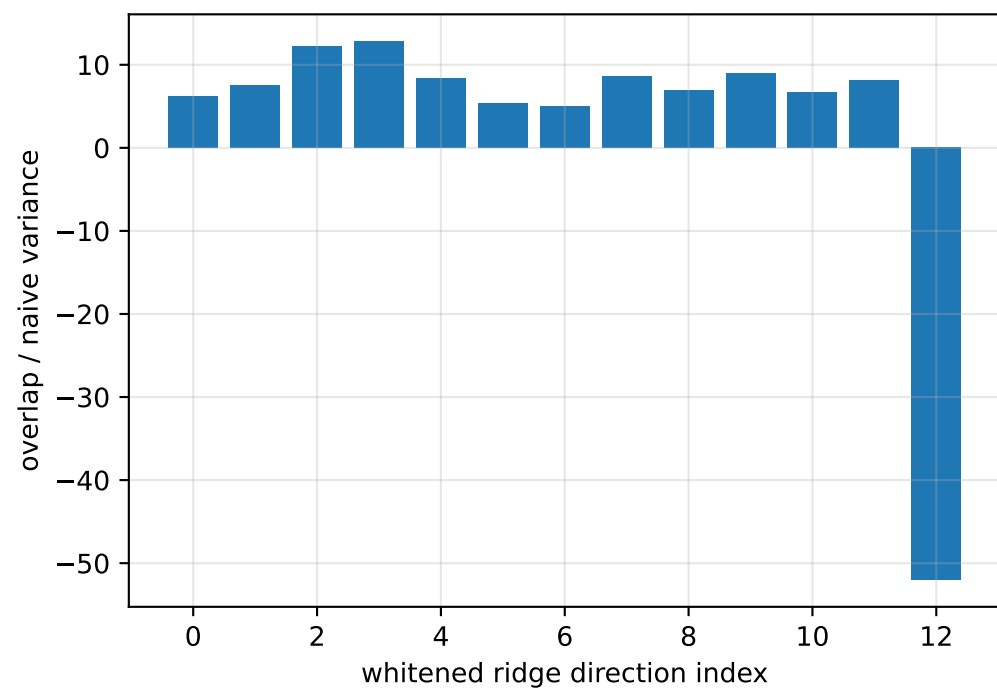
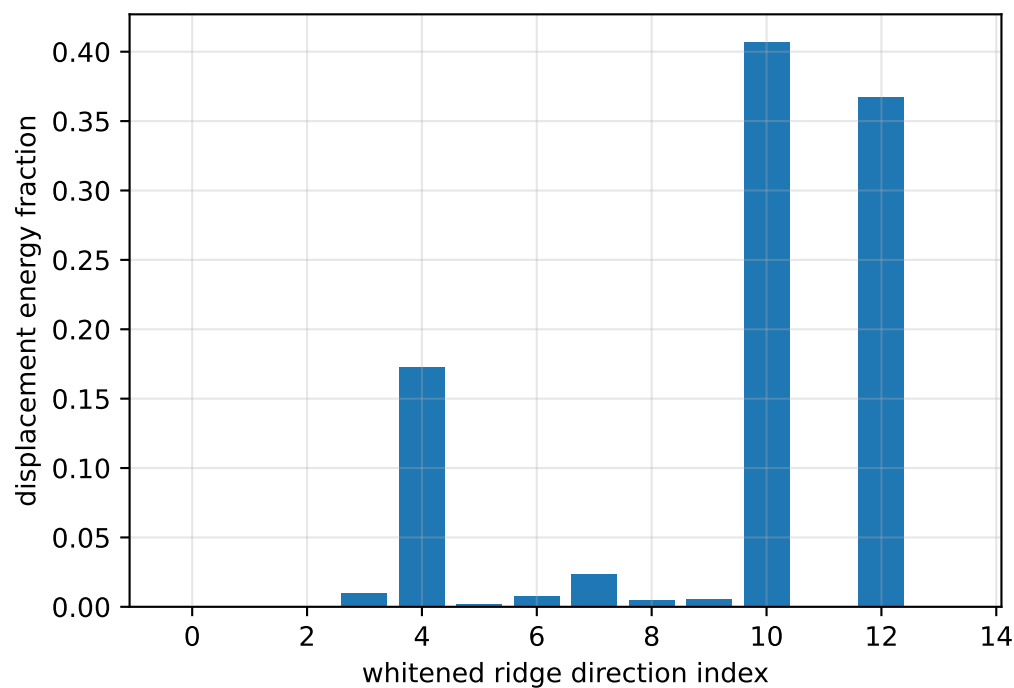
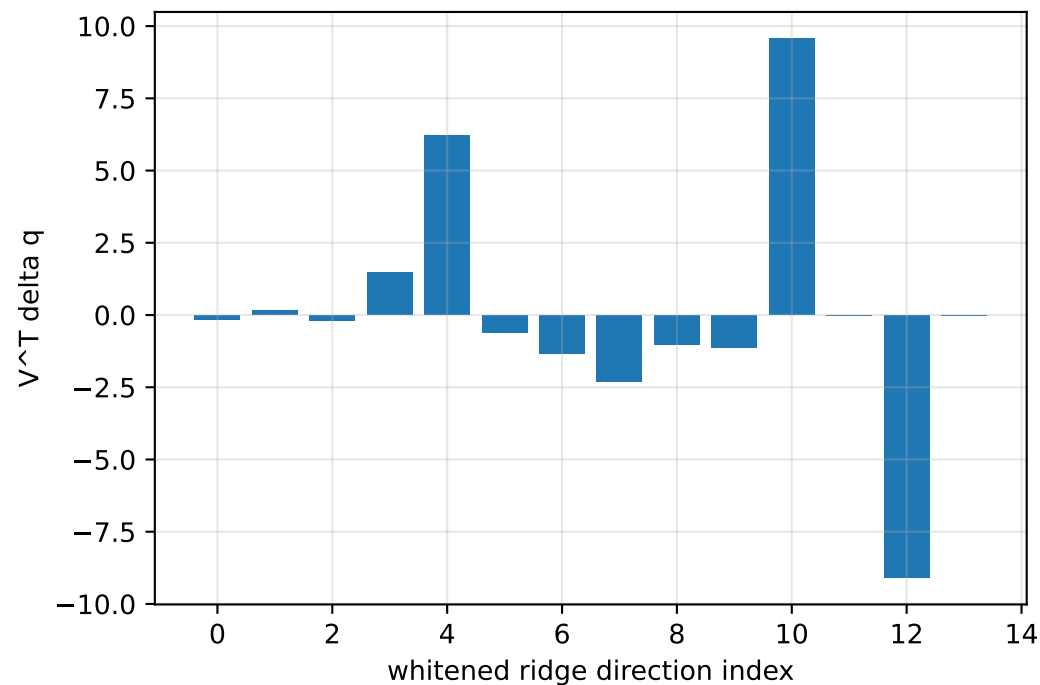
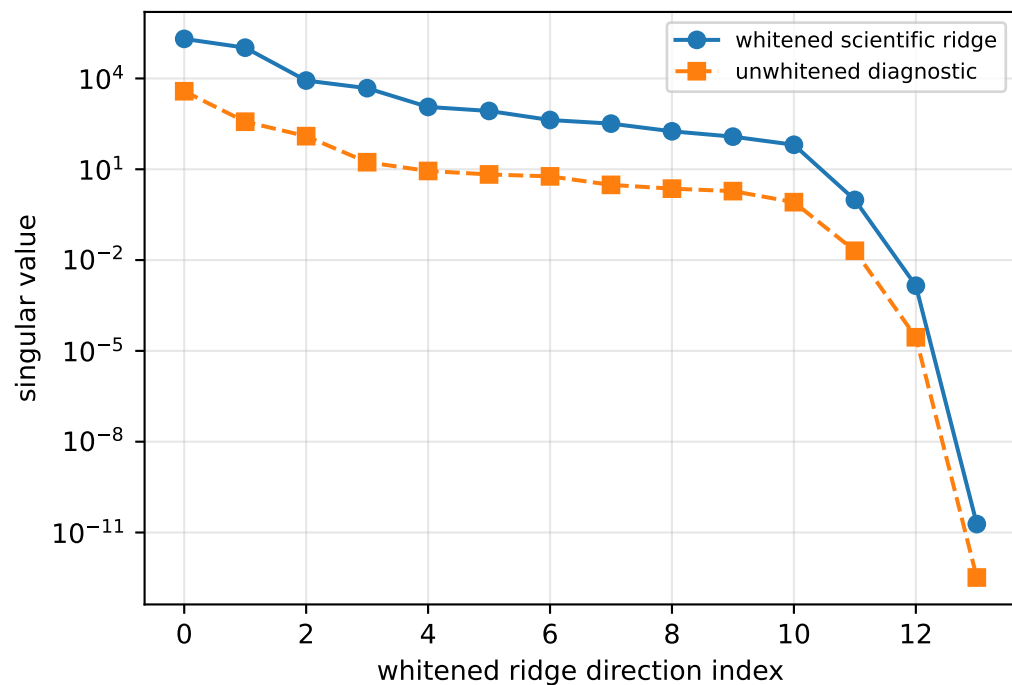
rotor lag [s] 0.214817708

gimbal lag [s] 0.0460151961

sweep_lag_bounds_000: covariance weighting

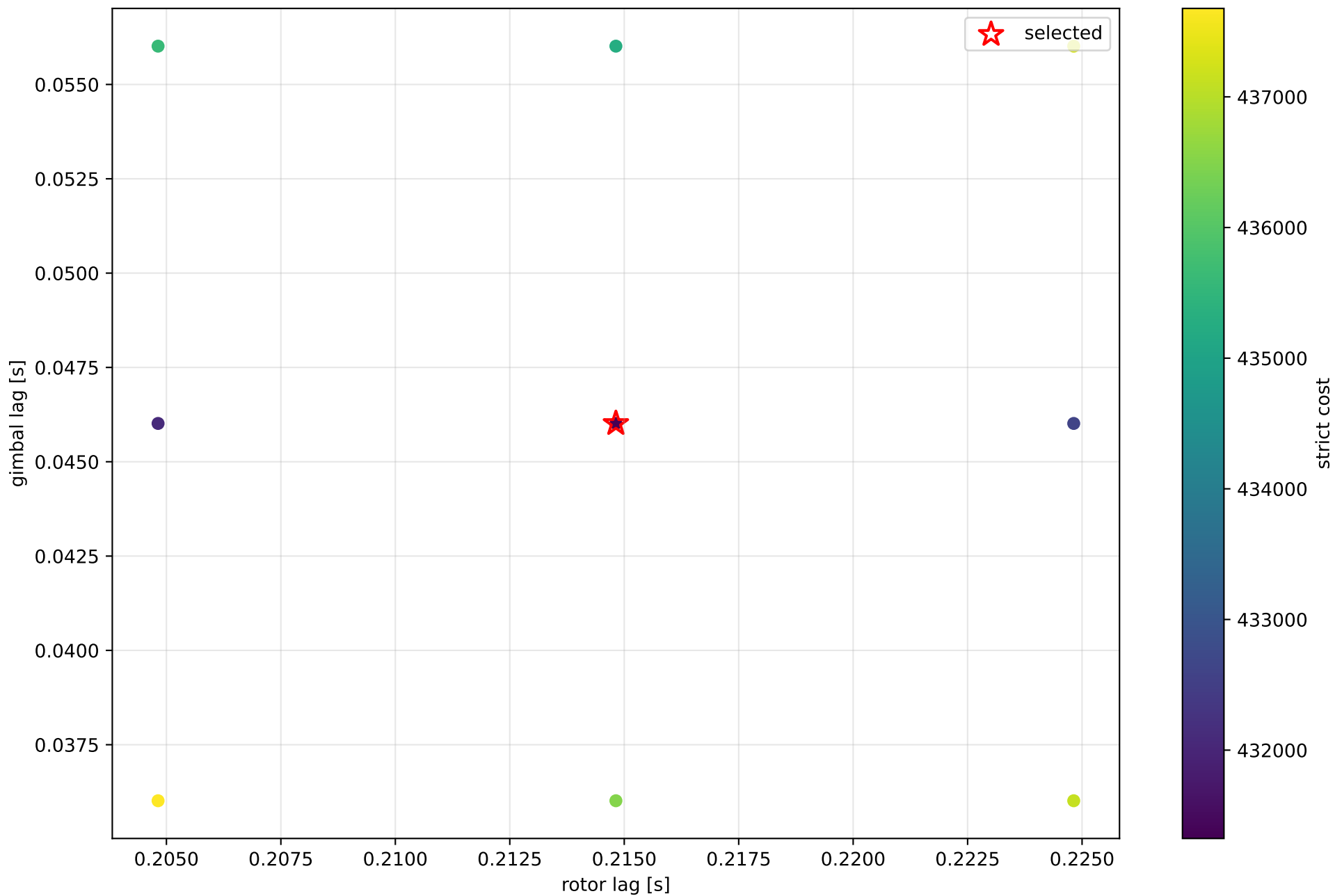


sweep_lag_bounds__000: parameter ridge



exact scale gauge: $\|v_{\text{scale}}\|=7.43746\text{e-}11$, rank=13, nullity=1

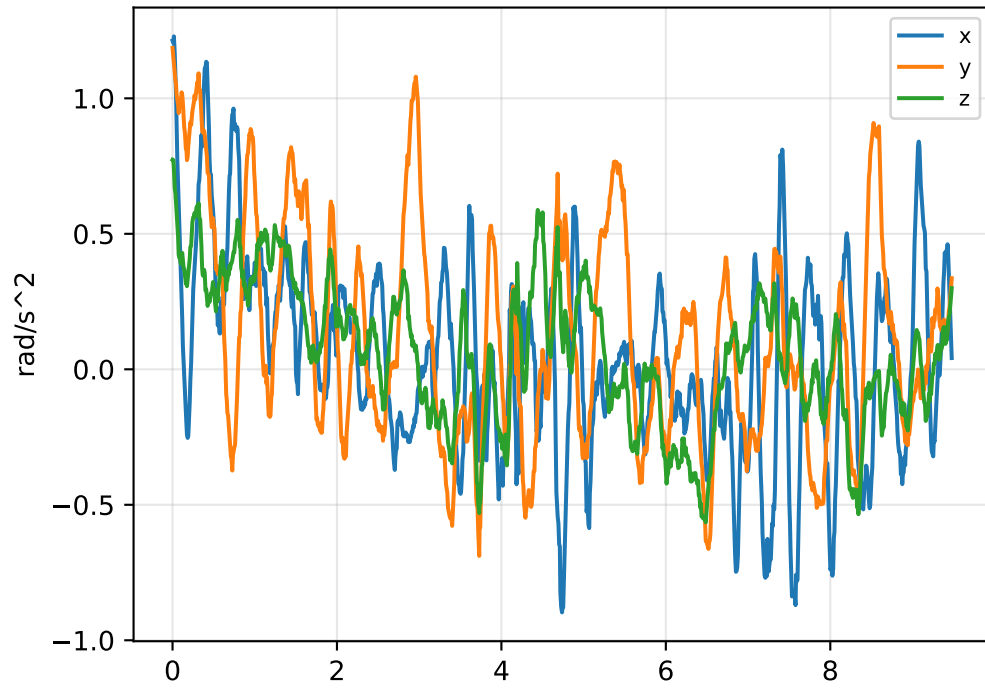
sweep_lag_bounds__000: strict lag candidates



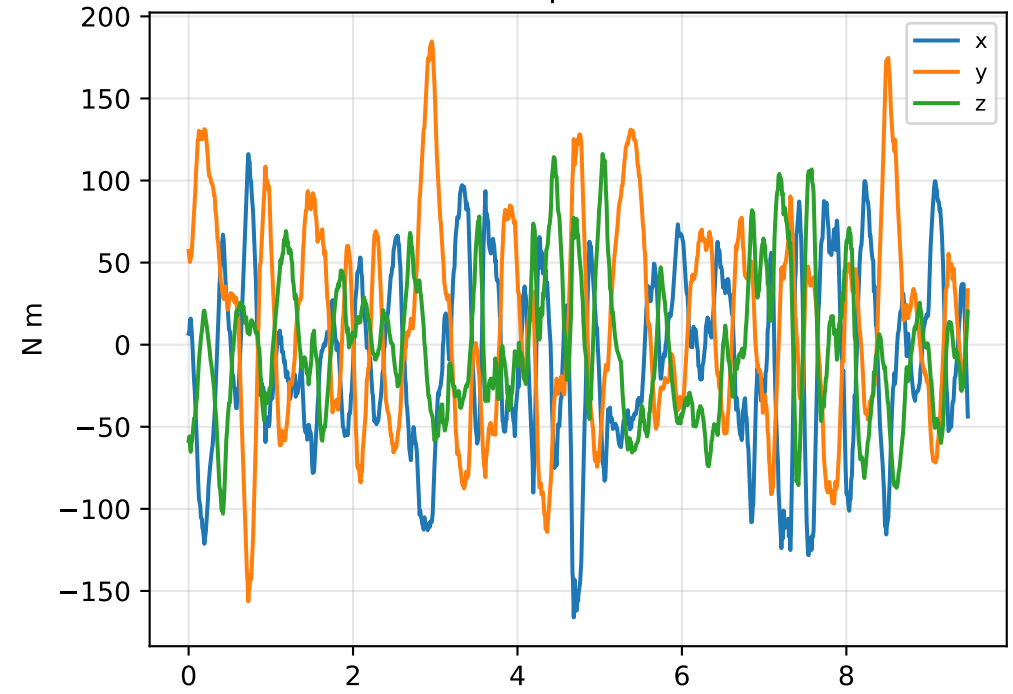
boundary flags: rotor lower=False upper=False; gimbal lower=False upper=False

sweep_lag_bounds_000: acceleration/wrench closure

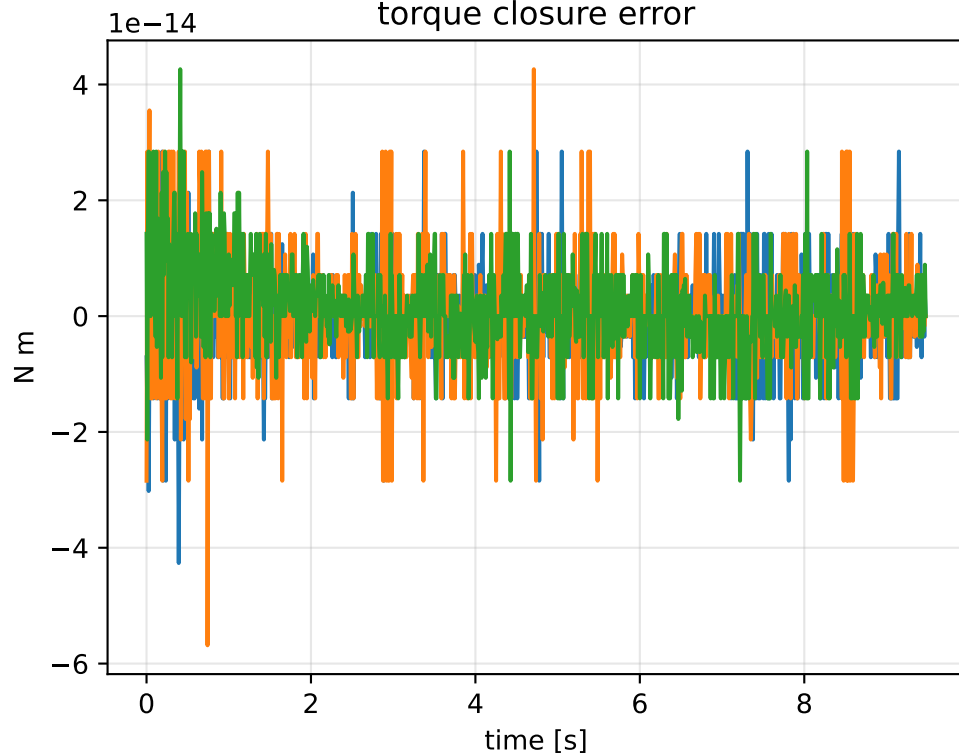
angular-acceleration residual



raw torque residual



torque closure error



```
principal moments [kg m^2]
nominal: [0.064953 0.065 0.128992]
estimated: [9.254903e-02 2.225780e+02 2.226705e+02]
estimated / nominal: [1.424869e+00 3.424273e+03 1.726234e+03]

force closure max/rms: 3.38618e-15 / 6.9887e-16
torque closure max/rms: 5.68434e-14 / 8.33017e-15
```